

WORKED EXAMPLES

PART -A

Find $\frac{dy}{dx}$ if

1) $y = (2x + 5)^3$

2) $y = \sqrt{\sin x}$

3) $y = \cos^4 x$

4) $y = e^{\tan x}$

5) $y = \log (\sec x)$

6) $y = \sin mx$

7) $y = \sec \sqrt{x}$

8) $y = \cos (2 - 3x)$

Solution:

1) $y = (2x + 5)^3$

$y = u^3$ where $u = 2x + 5$

$$\frac{dy}{du} = 3u^2 \quad \left| \quad \frac{du}{dx} = 2 \right.$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= (3u^2) (2)$$

$$= 6 (2x + 5)^2$$

2) $y = \sqrt{\sin x}$

$y = \sqrt{u}$ where $u = \sin x$

$$\frac{dy}{du} = \frac{1}{2\sqrt{u}} \quad \left| \quad \frac{du}{dx} = \cos x \right.$$

$$\frac{dy}{dx} = \left(\frac{1}{2\sqrt{u}} \right) (\cos x)$$

$$= \frac{\cos x}{2\sqrt{\sin x}}$$

3) $y = \cos^4 x$

$y = u^4$ where $u = \cos x$

$$\frac{dy}{du} = 4u^3 \quad \left| \quad \frac{du}{dx} = -\sin x \right.$$

$$\frac{dy}{dx} = 4u^3 (-\sin x)$$

$$= -4 \sin x \cos^3 x$$

4) $y = e^{\tan x}$

$y = e^u$ where $u = \tan x$

$$\frac{dy}{du} = e^u \quad \left| \quad \frac{du}{dx} = \sec^2 x \right.$$

$$\frac{dy}{dx} = (e^u) (\sec^2 x)$$

$$= e^{\tan x} \sec^2 x$$

5) $y = \log (\sec x)$

$$y = \log u \text{ where } u = \sec x$$

$$\frac{dy}{du} = \frac{1}{u} \quad \left| \quad \frac{du}{dx} = \sec x \tan x \right.$$

$$\begin{aligned}\frac{dy}{dx} &= \left(\frac{1}{u} \right) (\sec x \tan x) \\ &= \frac{1}{\sec x} (\sec x \tan x) \\ &= \tan x\end{aligned}$$

6) $y = \sin mx$

$$y = \sin u \text{ where } u = mx$$

$$\frac{dy}{du} = \cos u \quad \left| \quad \frac{du}{dx} = m \right.$$

$$\begin{aligned}\frac{dy}{dx} &= (\cos u) (m) \\ &= m \cos mx\end{aligned}$$

7) $y = \cos (2 - 3x)$

$$y = \cos u \text{ where } u = 2 - 3x$$

$$\frac{dy}{du} = -\sin u \quad \left| \quad \begin{aligned} \frac{du}{dx} &= 0 - 3(1) \\ &= -3 \end{aligned} \right.$$

$$\begin{aligned}\frac{dy}{dx} &= (-\sin u) (-3) \\ &= 3 \sin(2 - 3x)\end{aligned}$$

8) $y = \sec \sqrt{x}$

$$y = \sec u \text{ where } u = \sqrt{x}$$

$$\frac{dy}{du} = \sec u \cdot \tan u \quad \left| \quad \frac{du}{dx} = \frac{1}{2\sqrt{x}} \right.$$

$$\begin{aligned}\frac{dy}{dx} &= (\sec u \tan u) \left(\frac{1}{2\sqrt{x}} \right) \\ &= \frac{1}{2\sqrt{x}} \cdot \sec \sqrt{x} \cdot \tan \sqrt{x}\end{aligned}$$

Note:

1. $\frac{d}{dx} (e^{mx}) = me^{mx}$
2. $\frac{d}{dx} (\sin mx) = m \cos mx$
3. $\frac{d}{dx} (\cos mx) = -m \sin mx$
4. $\frac{d}{dx} (\tan mx) = m \sec^2 mx$

5. $\frac{d}{dx} (\cot mx) = -m \operatorname{cosec}^2 mx$
 6. $\frac{d}{dx} (\sec mx) = m \sec mx \tan mx$
 7. $\frac{d}{dx} (\operatorname{cosec} mx) = -m \operatorname{cosec} mx \cot mx$

PART – B

Differentiate the following w.r.t 'x'.

- 1) $(2x^2 - 3x + 1)^3$ 2) $\cos(e^{5x})$ 3) $e^{\sin^2 x}$ 4) $\log(\sec x + \tan x)$ 5) $y = \operatorname{cosec}^3(5x+1)$

Solution:

1) $y = (2x^2 - 3x + 1)^3$

$y = u^3$ where $u = 2x^2 - 3x + 1$

$$\frac{dy}{du} = 3u^2 \quad \left| \quad \frac{du}{dx} = 4x - 3 \right.$$

$$\frac{dy}{dx} = (3u^2)(4x - 3)$$

$$= 3(2x^2 - 3x + 1)^2(4x - 3)$$

2) Let $y = \cos(e^{5x})$

$y = \cos u$ where $u = e^{5x}$

$$\frac{dy}{du} = -\sin u \quad \left| \quad \frac{du}{dx} = 5e^{5x} \right.$$

$$\frac{dy}{dx} = (-\sin u)(5e^{5x})$$

$$= -5e^{5x} \sin(e^{5x})$$

3) Let $y = e^{\sin^2 x}$

$y = e^u$ where $u = \sin^2 x$

$$\frac{dy}{du} = e^u \quad \left| \quad u = v^2 \text{ where } v = \sin x \right.$$

$$\frac{du}{dv} = 2v \quad \left| \quad \frac{dv}{dx} = \cos x \right.$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$$

$$= (e^u)(2v)(\cos x)$$

$$= 2e^{\sin^2 x} \sin x \cos x$$

$$= \sin 2x \cdot e^{\sin^2 x}$$

4) Let $y = \log(\sec x + \tan x)$

$y = \log u$ where $u = \sec x + \tan x$

$$\frac{dy}{du} = \frac{1}{u} \quad \left| \quad \frac{du}{dx} = \sec x \cdot \tan x + \sec^2 x \right.$$

$$= \sec x (\tan x + \sec x)$$

$$\frac{dy}{dx} = \left(\frac{1}{u} \right) [\sec x (\tan x + \sec x)]$$

$$= \frac{1}{(\sec x + \tan x)} \cdot \sec x (\tan x + \sec x)$$

$$= \sec x$$

5) Let $y = \operatorname{cosec}^3(5x+1)$

$y = u^3$ where $u = \operatorname{cosec}(5x+1)$

$$\frac{dy}{du} = 3u^2 \quad \left| \quad u = \operatorname{cosec} v \text{ where } v = 5x+1 \right.$$

$$\frac{du}{dv} = -\operatorname{cosec} v \cot v \quad \left| \quad \frac{dv}{dx} = 5 \right.$$

$$\frac{dy}{dx} = (3u^2)(-\operatorname{cosec} v \cot v)(5)$$

$$= -15 \operatorname{cosec}^2(5x+1) \cdot \operatorname{cosec}(5x+1) \cot(5x+1)$$

$$= -15 \operatorname{cosec}^3(5x+1) \cot(5x+1)$$

PART – C

Differentiate the following w.r.t x.

1) $y = \sin(e^x \log x)$

2) $(x^2 + 5)e^{-2x} \sin x$

3) $y = (x \cos x)^3$

4) $y = \log\left(\frac{1 - \cos x}{1 + \cos x}\right)$

5) $y = \sqrt{\frac{1+x^3}{1-x^3}}$

6) $y = e^{4x} + \sin(x^2 + 5)$

7) $y = x^3 e^{-5x} \log(\sec x)$

Solution:

1) $y = \sin(e^x \log x)$

$y = \sin u$ where $u = e^x \log x$

$$\frac{dy}{du} = \cos u \quad \left| \quad \begin{aligned} \frac{du}{dx} &= e^x \frac{1}{x} + \log x \times e^x \\ &\text{(using product rule)} \\ &= e^x \left(\frac{1}{x} + \log x \right) \end{aligned} \right.$$

$$\frac{dy}{dx} = (\cos u) \left[e^x \left(\frac{1}{x} + \log x \right) \right]$$

$$= \cos(e^x \log x) \left[e^x \left(\frac{1}{x} + \log x \right) \right]$$

2) $y = (x^2 + 5)e^{-2x} \sin x$

$$\begin{aligned} u &= x^2 + 5 & v &= e^{-2x} & w &= \sin x \\ \frac{du}{dx} &= 2x & \frac{dv}{dx} &= e^{-2x}(-2) & \frac{dw}{dx} &= \cos x \\ & & &= -2e^{-2x} & \end{aligned}$$

$$\begin{aligned}\frac{dy}{dx} &= u v \frac{dw}{dx} + v w \frac{du}{dx} + u w \frac{dv}{dx} \\ &= (x^2 + 5)e^{-2x} \cos x + e^{-2x} \sin x(2x) + (x^2 - 5)\sin x(-2e^{-2x})\end{aligned}$$

$$3) y = (x \cos x)^3$$

$$y = u^3 \text{ where } u = x \cos x$$

$$\frac{dy}{du} = 3u^2 \quad \left| \begin{array}{l} \frac{du}{dx} = x(-\sin x) + \cos x(1) \\ \text{(using product rule)} \\ = -x \sin x + \cos x \end{array} \right.$$

$$\begin{aligned}\frac{dy}{dx} &= (3u^2)(-x \sin x + \cos x) \\ &= 3(x \cos x)^2(-x \sin x + \cos x)\end{aligned}$$

$$4) y = \log \left(\frac{1 - \cos x}{1 + \cos x} \right)$$

$$y = \log u \text{ where } u = \frac{1 - \cos x}{1 + \cos x}$$

$$\frac{dy}{du} = \frac{1}{u} \quad \left| \begin{array}{l} \frac{du}{dx} = \frac{(1 + \cos x)(+\sin x) - (1 - \cos x)(-\sin x)}{(1 + \cos x)^2} \\ \text{(using Quotient rule)} \end{array} \right.$$

$$\begin{aligned}\frac{dy}{dx} &= \left(\frac{1}{u} \right) \left[\frac{\sin x(1 + \cos x) + \sin x(1 - \cos x)}{(1 + \cos x)^2} \right] \\ &= \left(\frac{1 + \cos x}{1 - \cos x} \right) \left[\frac{2 \sin x}{(1 + \cos x)^2} \right] \\ &= \frac{2 \sin x}{(1 + \cos x)(1 - \cos x)} = \frac{2 \sin x}{1 - \cos^2 x} \\ &= \frac{2 \sin x}{\sin^2 x} \\ &= \frac{2}{\sin x}\end{aligned}$$

$$5) y = \sqrt{\frac{1 + x^3}{1 - x^3}}$$

$$\begin{array}{l} y = \sqrt{u} \\ \frac{dy}{du} = \frac{1}{2\sqrt{u}} \end{array} \quad \left| \begin{array}{l} \text{where } u = \frac{1 + x^3}{1 - x^3} \\ \frac{du}{dx} = \frac{(1 + x^3)(-3x^2) - (1 - x^3)(3x^2)}{(1 - x^3)^2} \\ = 3x^2 \left[\frac{-(1 + x^3) - (1 - x^3)}{(1 - x^3)^2} \right] \\ = \frac{-6x^2}{(1 - x^3)^2} \end{array} \right.$$

$$\begin{aligned}
\frac{dy}{dx} &= \left(\frac{1}{2\sqrt{u}} \right) \frac{-6x^2}{(1+x^3)^2} \\
&= \frac{1}{2\sqrt{\frac{1-x^3}{1+x^3}}} \left[\frac{-6x^2}{(1+x^3)^2} \right] \\
&= \frac{-3x^2}{(1+x^3)^2} \sqrt{\frac{1+x^3}{1-x^3}}
\end{aligned}$$

$$6) y = e^{4x} + \sin(x^2 + 5)$$

$$\begin{aligned}
\frac{dy}{dx} &= e^{4x} \times 4 + \cos(x^2 + 5) \times (2x + 0) \\
&= 4e^{4x} + 2x \cos(x^2 + 5)
\end{aligned}$$

$$7) y = x^3 e^{-5x} \log(\sec x)$$

$$\begin{array}{l|l|l}
u = x^3 & v = e^{-5x} & w = \log(\sec x) \\
\frac{du}{dx} = 3x^2 & \frac{dv}{dx} = e^{-5x}(-5) & \frac{dw}{dx} = \frac{1}{\sec x} \sec x \cdot \tan x \\
& = -5e^{-5x} & = \tan x
\end{array}$$

$$\begin{aligned}
\frac{dy}{dx} &= uv \frac{dw}{dx} + vw \frac{du}{dx} + uw \frac{dv}{dx} \\
&= x^3 e^{-5x} \tan x + e^{-5x} \log(\sec x)(3x^2) + x^3 \log(\sec x)(-5e^{-5x})
\end{aligned}$$

5.1.2 DIFFERENTIATION OF INVERSE TRIGONOMETRIC FUNCTIONS

If $x = \sin y$ then we get a unique real value of x to the given angle y . But if x is given y has many values.

We can express y in terms of x as $y = \sin^{-1}x$ "Inverse sine of x ". The Quantities $\sin^{-1}x$, $\cos^{-1}x$, $\tan^{-1}x$, $\cot^{-1}x$, $\sec^{-1}x$, $\operatorname{cosec}^{-1}x$ are called Inverse Trigonometric Functions.

Differential co-efficient of $\sin^{-1}x$:

$$\text{Let } y = \sin^{-1}x$$

$$\Rightarrow \sin y = x$$

Differentiating w.r.t x

$$\begin{aligned}
\cos y \cdot \frac{dy}{dx} &= 1 \\
\frac{dy}{dx} &= \frac{1}{\cos y} \\
&= \frac{1}{\sqrt{1 - \sin^2 y}}
\end{aligned}$$

$$\text{i.e., } \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1 - x^2}}$$

Similarly we get

$$(i) \frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$$

$$(ii) \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$(iii) \frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$$

$$(iv) \frac{d}{dx}(\sec^{-1} x) = \frac{-1}{x\sqrt{x^2-1}}$$

$$(v) \frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{-1}{x\sqrt{x^2-1}}$$

WORKED EXAMPLES

PART – A

Find $\frac{dy}{dx}$ if

$$(i) y = \sin^{-1} \sqrt{x}$$

$$(ii) y = \cos^{-1}(2x)$$

$$(iii) y = \tan^{-1}(\log x)$$

Solution:

$$(i) y = \sin^{-1} \sqrt{x}$$

$$\frac{dy}{dx} = \sin^{-1} u \text{ where } u = \sqrt{x}$$

$$\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}} \quad \left| \quad \frac{du}{dx} = \frac{1}{2\sqrt{x}} \right.$$

$$\frac{dy}{dx} = \left(\frac{1}{\sqrt{1-u^2}} \right) \left(\frac{1}{2\sqrt{x}} \right)$$

$$= \frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{1}{2\sqrt{x}}$$

$$= \frac{1}{\sqrt{1-x}} \cdot \frac{1}{2\sqrt{x}}$$

$$= \frac{1}{2\sqrt{x(1-x)}}$$

$$(ii) y = \cos^{-1}(2x)$$

$$y = \cos^{-1} u \text{ where } u = 2x$$

$$\frac{dy}{du} = \frac{-1}{\sqrt{1-u^2}} \quad \left| \quad \frac{du}{dx} = 2 \right.$$

$$\frac{dy}{dx} = \frac{-1}{\sqrt{1-u^2}} \cdot 2$$

$$= \frac{-1}{\sqrt{1-(2x)^2}} \cdot 2$$

$$\frac{dy}{dx} = \frac{-2}{\sqrt{1-4x^2}}$$

(iii) $y = \tan^{-1}(\log x)$

$y = \tan^{-1}u$ where $u = \log x$

$$\frac{dy}{du} = \frac{1}{1+u^2} \quad \left| \quad \frac{du}{dx} = \frac{1}{x} \right.$$

$$\begin{aligned} \frac{dy}{dx} &= \left(\frac{1}{1+u^2} \right) \left(\frac{1}{x} \right) \\ &= \frac{1}{x[1+(\log x)^2]} \end{aligned}$$

PART – B

Find $\frac{dy}{dx}$ if

(i) $y = \sin^{-1}(\cos x)$ (ii) $y = \cos^{-1}(2x + 3)$ (iii) $y = \cos^{-1}(2x^2 - 1)$ (iv) $y = \sin^{-1}(3x - 4x^3)$

Solution:

(i) $y = \sin^{-1}(\cos x)$

$y = \sin^{-1}u$ where $u = \cos x$

$$\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}} \quad \left| \quad \frac{du}{dx} = -\sin x \right.$$

$$\begin{aligned} \frac{dy}{dx} &= \left(\frac{1}{\sqrt{1-u^2}} \right) (-\sin x) \\ &= \frac{1}{\sqrt{1-\cos^2 x}} (-\sin x) \\ &= \frac{-\sin x}{\sqrt{\sin^2 x}} = -1 \end{aligned}$$

(ii) $y = \cos^{-1}(2x + 3)$

$$\begin{aligned} \frac{dy}{dx} &= \frac{-1}{\sqrt{1-(2x+3)^2}} \cdot \frac{d}{dx}(2x+3) \\ &= \frac{-1}{\sqrt{1-(2x+3)^2}} \times 2 \\ &= \frac{-2}{\sqrt{1-(2x+3)^2}} \end{aligned}$$

(iii) $y = \cos^{-1}(2x^2 - 1)$

$$\begin{aligned} \frac{dy}{dx} &= \frac{-1}{\sqrt{1-(2x^2-1)^2}} \cdot \frac{d}{dx}(2x^2) \\ &= \frac{-1}{\sqrt{1-(2x^2-1)^2}} \cdot 4x \\ &= \frac{-4x}{\sqrt{1-(2x^2-1)^2}} = \frac{-4x}{\sqrt{1-4x^4+4x^2-1}} = \frac{-4x}{\sqrt{4x^2-4x^4}} = \frac{-4x}{2x\sqrt{1-x^2}} = \frac{-2}{\sqrt{1-x^2}} \end{aligned}$$

ALITER

$$y = \cos^{-1}(2x^2 - 1)$$

$$\text{Put } x = \cos \theta$$

$$y = \cos^{-1}(2 \cos^2 \theta - 1)$$

$$= \cos^{-1}(\cos 2\theta)$$

$$y = 2\theta$$

$$y = 2 \cos^{-1} x \quad x = \cos \theta \Rightarrow \theta = \cos^{-1} x$$

$$\frac{dy}{dx} = 2 \times \frac{-1}{\sqrt{1-x^2}} = \frac{-2}{\sqrt{1-x^2}}$$

$$(iv) y = \sin^{-1}(3x - 4x^3)$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-(3x-4x^3)^2}} \cdot (3-12x^2)$$

ALITER

$$\text{Put } x = \sin \theta$$

$$y = \sin^{-1}(3 \sin \theta - 4 \sin^3 \theta)$$

$$= \sin^{-1}(\sin 3\theta)$$

$$y = 3\theta$$

$$y = 3 \sin^{-1} x \quad x = \sin \theta \Rightarrow \theta = \sin^{-1} x$$

$$\frac{dy}{dx} = \frac{3}{\sqrt{1-x^2}}$$

PART – C

$$(i) y = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$$

$$(ii) y = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$$

$$(iii) y = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$$

$$(iv) y = (1+x^2) \tan^{-1} x$$

Solution:

$$(i) y = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$$

$$\text{Let } x = \tan \theta \Rightarrow \theta = \tan^{-1} x$$

$$y = \cos^{-1} \left(\frac{1-\tan^2 \theta}{1+\tan^2 \theta} \right)$$

$$= \cos^{-1}(\cos 2\theta)$$

$$= 2\theta$$

$$y = 2 \tan^{-1} x$$

$$\frac{dy}{dx} = 2 \frac{1}{1+x^2}$$

$$(ii) y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$$

$$\text{Let } x = \tan \theta \Rightarrow \theta = \tan^{-1}x$$

$$y = \sin^{-1}\left(\frac{2 \tan \theta}{1 + \tan^2 \theta}\right)$$

$$= \sin^{-1}(\sin 2\theta)$$

$$= 2\theta$$

$$y = 2 \tan^{-1} x$$

$$\frac{dy}{dx} = 2 \cdot \frac{1}{1+x^2}$$

$$(iii) y = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

$$\text{Let } x = \tan \theta \Rightarrow \theta = \tan^{-1}x$$

$$y = \tan^{-1}\left(\frac{2 \tan \theta}{1 - \tan^2 \theta}\right)$$

$$= \tan^{-1}(\tan 2\theta)$$

$$y = 2\theta$$

$$y = 2 \tan^{-1} x$$

$$\frac{dy}{dx} = 2 \cdot \frac{1}{1+x^2}$$

$$(iv) y = (1+x^2) \tan^{-1} x$$

$$\frac{dy}{dx} = (1+x^2) \frac{1}{1+x^2} + \tan^{-1} x (0+2x) \quad (\text{using product rule})$$

$$= 1 + 2x \tan^{-1} x$$

5.1.3 IMPLICIT FUNCTION

If the variables x and y are connected by a relation $f(x, y) = 0$ then it is called an implicit function.

i.e If the dependent variable (y) cannot be expressed explicitly in terms of the independent variable x .

Example: (i) $x^2 + y^2 - 4x + y + 12 = 0$

(ii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are implicit function

PART –B

1. Find $\frac{dy}{dx}$ if

(i) $x^2 + y^2 + 2y = 0$

(ii) $x^2 + y^2 = a^2$

(iii) $xy = c^2$

(iv) $\sqrt{y} + \sqrt{x} = \sqrt{a}$

Solution:

(i) $x^2 + y^2 + 2y = 0$

Differentiate w.r.t x

$$2x + 2y \frac{dy}{dx} + 2 \frac{dy}{dx} = 0$$

$$x + y \frac{dy}{dx} + \frac{dy}{dx} = 0$$

$$(y+1) \frac{dy}{dx} = -x$$

$$\frac{dy}{dx} = \frac{-x}{1+y}$$

(ii) $x^2 + y^2 = a^2$

Differentiate w.r.t x

$$2x + 2y \frac{dy}{dx} = 0$$

$$x + y \frac{dy}{dx} = 0$$

$$y \frac{dy}{dx} = -x$$

$$\frac{dy}{dx} = \frac{-x}{y}$$

(iii) $xy = c^2$

Differentiate w.r.t x

$$x \cdot \frac{dy}{dx} + y(1) = 0 \text{ (using product rule)}$$

$$x \frac{dy}{dx} = -y$$

$$\frac{dy}{dx} = \frac{-y}{x}$$

(iv) $\sqrt{y} + \sqrt{x} = \sqrt{a}$

Differentiate w.r.t x

$$\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \cdot \frac{dy}{dx} = 0$$

Multiply by 2

$$\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{y}} \cdot \frac{dy}{dx} = 0$$

$$\frac{1}{\sqrt{y}} \frac{dy}{dx} = \frac{-1}{\sqrt{x}}$$

$$\frac{dy}{dx} = \frac{-\sqrt{y}}{\sqrt{x}}$$

PART – C

1. Find $\frac{dy}{dx}$ if

(i) $\cos x + \sin y = c$ (ii) $x^2 \sin y = c$ (iii) $y = a + xe^y$

Solution:

(i) $\cos x + \sin y = c$

Differentiate w.r.t x

$$-\sin x + \cos y \cdot \frac{dy}{dx} = 0$$

$$\cos y \cdot \frac{dy}{dx} = \sin x$$

$$\frac{dy}{dx} = \frac{\sin x}{\cos y}$$

(ii) $x^2 \sin y = c$

Differentiate w.r.t x

$$x^2 \cos y \cdot \frac{dy}{dx} + \sin y (2x) = 0 \quad (\text{using product rule})$$

$$x^2 \cos y \cdot \frac{dy}{dx} = -2x \sin y$$

$$\frac{dy}{dx} = \frac{-2x \sin y}{x^2 \cos y}$$

$$\frac{dy}{dx} = \frac{-2}{x} \tan y$$

(iii) $y = a + xe^y$

Differentiate w.r.t x

$$\frac{dy}{dx} = 0 + x e^y \cdot \frac{dy}{dx} + e^y$$

$$\frac{dy}{dx} - x e^y \frac{dy}{dx} = e^y$$

$$\frac{dy}{dx} (1 - x e^y) = e^y$$

$$\frac{dy}{dx} = \frac{e^y}{1 - x e^y}$$

2. Find $\frac{dy}{dx}$ if

(i) $x^3 + y^3 = 3axy$ (ii) $x^2 + y^2 - 4x + 6y - 5 = 0$ (iii) $x^2 + y^2 = xy$

(iv) $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

(i) $x^3 + y^3 = 3axy$

Differentiate w.r.t x

$$3x^2 + 3y^2 \frac{dy}{dx} = 3a \left[x \frac{dy}{dx} + y(1) \right]$$

Divide by 3

(using product rule)

$$x^2 + y^2 \frac{dy}{dx} = a \left[x \frac{dy}{dx} + y \right]$$

$$y^2 \frac{dy}{dx} - ax \frac{dy}{dx} = ay - x^2$$

$$(y^2 - ax) \frac{dy}{dx} = ay - x^2$$

$$\frac{dy}{dx} = \frac{ay - x^2}{y^2 - ax}$$

(ii) $x^2 + y^2 - 4x + 6y - 5 = 0$

Differentiating w.r.t x

$$2x + 2y \frac{dy}{dx} - 4 + 6 \frac{dy}{dx} = 0$$

$$2y \frac{dy}{dx} + 6 \frac{dy}{dx} = -2x + 4$$

$$\frac{dy}{dx} (2y + 6) = -2x + 4$$

$$\frac{dy}{dx} = \frac{-2x + 4}{2y + 6} = \frac{2(-x + 2)}{2(y + 3)}$$

$$= \frac{-x + 2}{y + 3}$$

(iii) $x^2 + y^2 = xy$

uv rule

Differentiating w.r.t x,

$$2x + 2y \frac{dy}{dx} = x \frac{dy}{dx} + y(1)$$

$$2y \frac{dy}{dx} - x \frac{dy}{dx} = y - 2x$$

$$\frac{dy}{dx} (2y - x) = y - 2x$$

$$\frac{dy}{dx} = \frac{y - 2x}{2y - x}$$

(iv) $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

Differentiate w.r.t x

$$2ax + 2h \left[x \frac{dy}{dx} + y(1) \right] + b(2y) \frac{dy}{dx} + 2g(1) + 2f \frac{dy}{dx} + 0 = 0$$

Divide by 2

$$ax + hx \frac{dy}{dx} + hy + by \frac{dy}{dx} + g + f \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} (hx + by + f) = -ax - hy - g$$

$$\frac{dy}{dx} = \frac{-ax - hy - g}{hx + by + f}$$

5.2.1 SUCCESSIVE DIFFERENTIATION

If $y = f(x)$ then $\frac{dy}{dx} = f'(x)$ is a function of x or constant which can be differentiated once again.

$\frac{dy}{dx} = f'(x)$ is the first order derivative of y w.r.t x .

$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$ is the second order derivative of y w.r.t x

i.e the derivate of $\frac{dy}{dx}$ w.r.t x .

$\frac{d^3y}{dx^3} = \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right)$ is the third order derivative of y w.r.t x and so on.

Notations of successive derivatives

$$\frac{dy}{dx} = y_1 = y' = f'(x) = D(y)$$

$$\frac{d^2y}{dx^2} = y_2 = y'' = f''(x) = D^2(y)$$

$$\frac{d^ny}{dx^n} = y_n = y^{(n)} = f^{(n)}(x) = D^n(y)$$

WORKED EXAMPLES

PART – A

1. If $y = e^{3x}$ find $\frac{d^2y}{dx^2}$.

Solution:

$$\frac{dy}{dx} = 3e^{3x}$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= 3(3e^{3x}) \\ &= 9e^{3x} \end{aligned}$$

2. If $y = \sin 3x$ find $\frac{d^2y}{dx^2}$.

Solution:

$$\frac{dy}{dx} = 3 \cos 3x$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= 3(-3 \sin 3x) \\ &= -9 \sin 3x \end{aligned}$$